

Practical 2

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The screenshot shows the CodeTANTRA web interface. The top navigation bar includes the logo, home link, and user profile. The main content area is titled 'Practical 2' and features a sidebar with a course tree. The tree lists 'Practical 1' and 'Practical 2', with 'Practical 2' currently selected. Under 'Practical 2', there are links for 'Practice Lab Assignment' and 'Lab Assignment', both marked as 'Unit - 100% completed'.

This screenshot displays the 'List operations' lab assignment. On the left, the problem description asks for a Python program with a menu-driven interface for managing a list of integers. The menu options are: 1. Add, 2. Remove, 3. Display, 4. Quit. The instructions specify how to handle invalid inputs, list removal, and display. On the right, the 'testOps.py' code editor shows a Python script implementing these operations. Below the code, the execution results are shown: '2 out of 2 shown test case(s) passed' and '1 out of 1 hidden test case(s) passed'. The test case details show the expected and actual outputs for the 'Add', 'Remove', 'Display', and 'Quit' operations, along with the input choices and the resulting list of integers.

Course

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2.1.2. Dictionary Operations

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.

2. Update – Update the value of an existing key using user input.

3. Deletion – Delete a specified key from the dictionary using user input.

4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.

2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.

3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.

4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

All operations must be performed using dictionary methods.

Perform deletion only if possible; leave the dictionary unchanged.

Refer to the visible test cases for better understanding and strictly match with the Input/outputs.

Sample Test Cases

DictOper...

```
1 # Initial dictionary with 10 predefined records
2 student = {
3     1: "Amit",
4     2: "Riya",
5     3: "Kiran",
6     4: "Neha",
7     5: "Arjun",
8     6: "Pooja",
9     7: "Rahul",
10    8: "Sneha",
11    9: "Vikram",
12    10: "Anjali"
13 }
14 print("Original Dictionary:", student)
15 key = int(input())
```

Average time0.018 sMaximum time0.024 s

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 124 msDebug

Expected outputOriginal Dictionary: (1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali')
After Insertion: (1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh')
After Insertion: (1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh')

TerminalTest cases

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Search

ENG IN

21:3503-05-2026

Course

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2.2.1. Linear search Technique

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

The first line of Input contains the array of Integers which are separated by space

The last line of Input contains the key element to be searched

Output format:

If the element is found, print the index. If the element found multiple times, print the starting index

If the element is not found, print Not found.

Sample Test Cases:

Input:
1 2 3 4 3 5 6
3
Output:
2

Sample Test Cases

CTP1/ER...

```
1 arr=list(map(int,input().split()))
2 key=int(input())
3 found=False
4 for i in range(len(arr)):
5     if arr[i]==key:
6         print(i)
7         found=True
8         break
9 if not found:
10    print("Not Found")
11
```

Average time0.009 sMaximum time0.011 s

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 111 msDebug

Expected output1 2 3 4 3 5 6
3
2

Actual output1 2 3 4 3 5 6
3
2

Test cases 24 ms

TerminalTest cases

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Search

ENG IN

21:3903-05-2026

The screenshot shows the CodeTANTRA website interface. On the left, the problem '23.2. Captain of the Team' is described: 'You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.' It specifies the input format (11 integers) and output format (the height of the tallest player). Below the problem description, there is a section for 'Sample Test Cases'.

On the right, the solution code is displayed in a Python editor. The code reads the input, splits it into a list of integers, and finds the maximum value using the `max()` function. The output is printed.

Below the code editor, the execution results are shown:

- Average time: 0.008 s
- Maximum time: 0.008 s
- 7.67 ms
- 8.00 ms
- 1 out of 1 shown test case(s) passed
- 2 out of 2 hidden test case(s) passed

A detailed view of 'Test case 1' shows the expected output and actual output, both being 171.